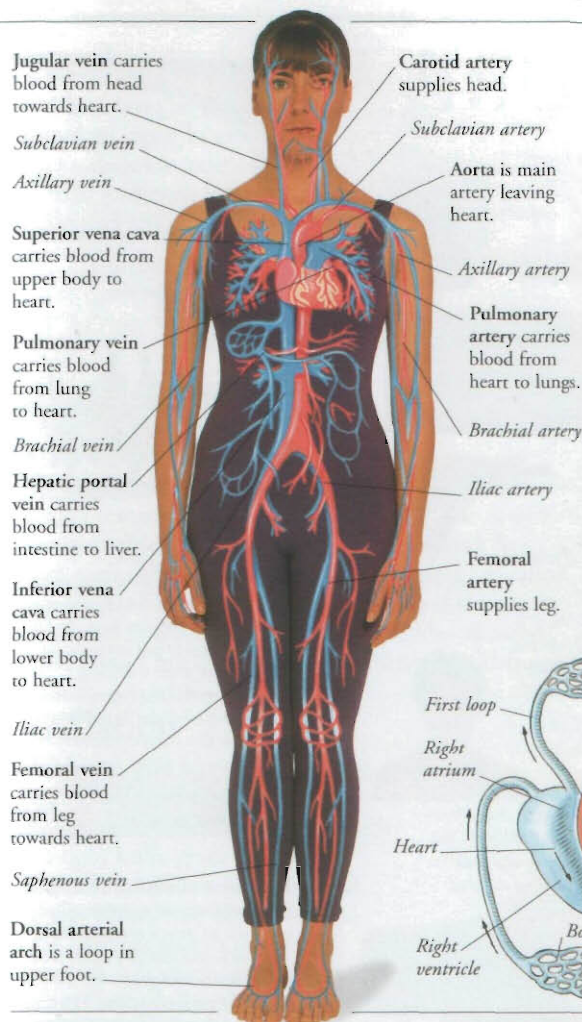


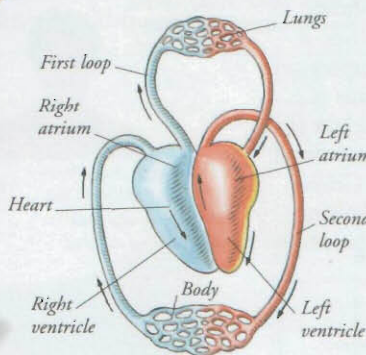


Circulatory system

The main blood vessel leaving the heart, the aorta, divides repeatedly, sending branches to major body regions. Blood returns to the heart along veins, which unite to form the superior and inferior venae cavae, the two major veins which re-enter the heart. A red blood cell takes just one minute to complete its journey around the circulatory system.

Digital artery supplies fingers with blood.

Digital vein carries blood away from fingers.



How blood circulates

There are, in fact, two parts to the circulatory system. The pulmonary circulation carries oxygen-poor blood from the right side of the heart along the pulmonary arteries to the lungs and back to the left side of the heart along the pulmonary veins. The systemic circulation carries oxygen-rich blood from the left side of the heart along the aorta to the body, and returns oxygen-poor blood to the right side of the heart.

Circulation in the arm
The blood vessels of the arm show how the circulatory system works. The brachial artery divides into several branches, including the radial artery. Veins carrying blood from the hand and wrist unite to form the brachial vein leaving the arm.

Axillary vein carries blood towards heart.

Axillary artery supplies arm with blood.

Cephalic vein

Basilic vein

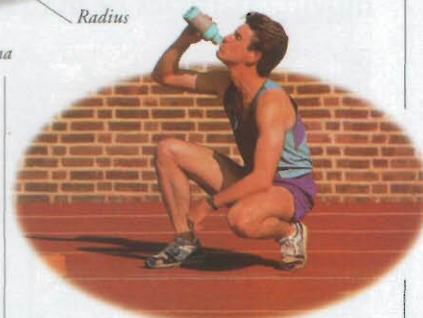
Brachial artery

Humerus

Radius

Ulna

Ulnar artery

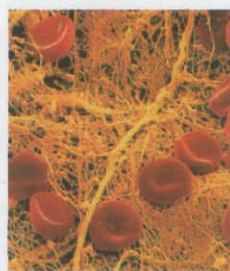
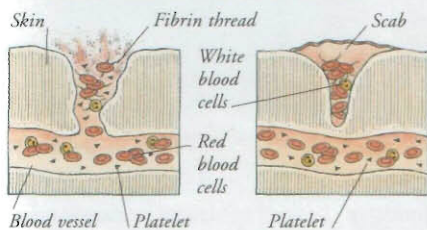


Flushing

Exercise can cause a reddening of the face and body known as flushing. This happens when, to cool the body down, blood vessels near the skin's surface widen as blood flow increases to lose heat from the skin. The increased blood flow makes the skin redder.

Blood clotting

When a blood vessel is damaged, clotting reduces the loss of blood. Platelets accumulate at the wound and stick together to form a plug. Red blood cells are trapped in threads of fibrin to form a clot. White blood cells prevent infection beneath a hard outer scab.



Fibrin

Chemicals in the blood and damaged cells trigger the production of the protein fibrin. This forms strands which trap red blood cells into a clot. A hard crust, called a scab, forms over the clot to protect the wound.

Blood vessels

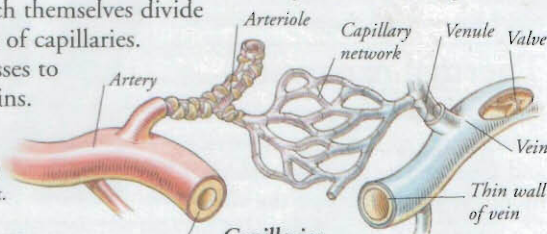
There are three types of blood vessels: arteries, veins, and capillaries. Arteries divide into smaller vessels called arterioles, which themselves divide into a network of capillaries. Blood then passes to venules and veins.

Arteries

Arteries carry blood away from the heart. They have thick, muscular walls that can withstand the high pressure produced when the heart beats. Arteries usually carry blood that is rich in oxygen.

Veins

Veins carry blood towards the heart. They have thin walls because pressure inside them is low. They contain valves to prevent blood flowing backwards.



Capillaries

Capillaries are the tiny blood vessels that carry blood between arterioles and venules. They supply individual cells with food and oxygen and remove wastes.

Karl Landsteiner

Austrian/American Karl Landsteiner (1868–1943) discovered the existence of blood groups, and made safe blood transfusion a reality. In 1900, Landsteiner showed that red blood cells may clump together when blood from different people is mixed. He worked out the ABO blood group system, and was awarded a Nobel Prize.



Blood groups

People belong to different blood groups, depending on the antigens (chemicals) in their red blood cells. The ABO blood group system has two antigens, A and B. It has four blood groups: A (carries A antigen); B (carries B antigen); AB (carries both antigens); and O (carries neither).

Regular blood donors are given a donor badge.



Blood compatibility

Blood transfusion is the donation of blood by one person to another. People who share the same blood groups can give or receive blood safely because their blood is compatible. In an emergency, however, people with type O blood can give blood to any other group.

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